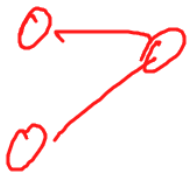
$y = mx + b$ → $\frac{1}{n}\sum(\hat{y}-y)^2$ = J = MSE

J = Cross Entropy = ??

N.N.



Perceptron

70%

> 0.5





Linear Regression

$\vec{X}$ $\vec{W}$

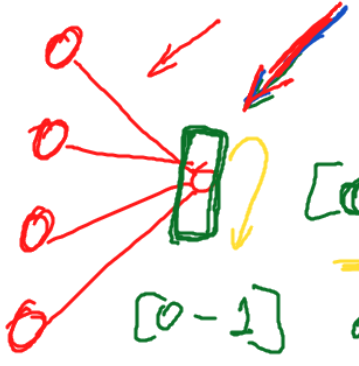


$[-\infty \; +\infty]$



Simplified (binary)

Logistic Regression



$[0-1]$ 0.5 $[0,1]$

N.N.

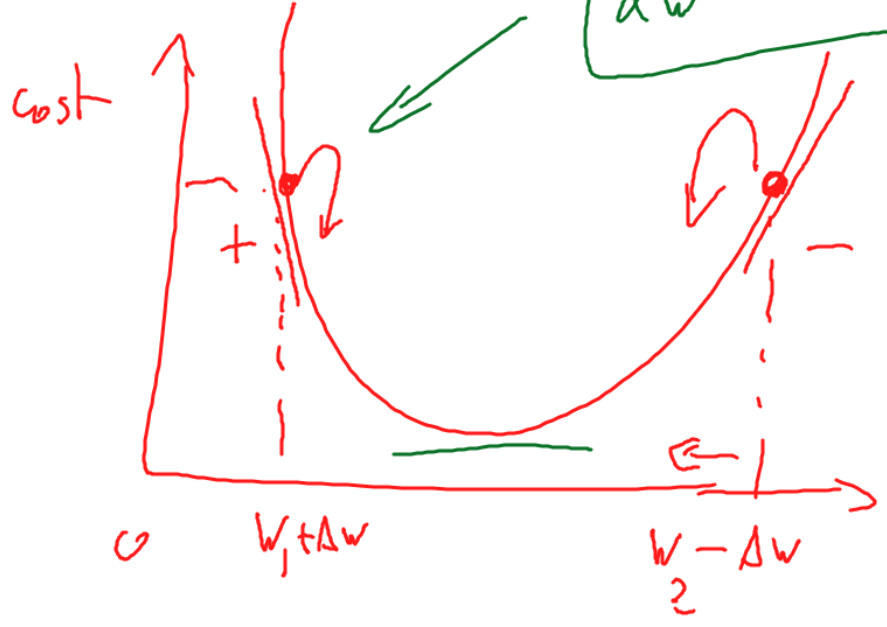
[Cat, Dog]
0 1

[Clr, ver, set]

Linear Regression

$$J_{MSE} = \frac{1}{n}\sum (\hat{y} - y)^2$$

$$\frac{dJ_{MSE}}{dw} =$$

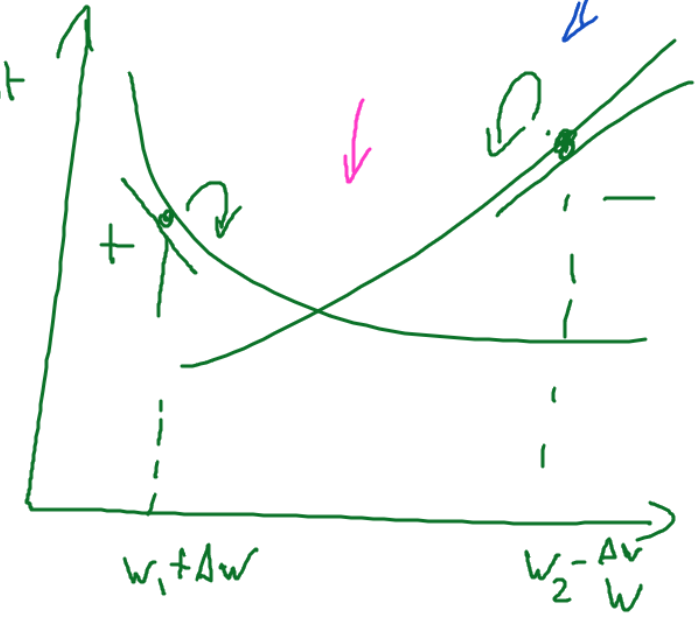


Logistic Regress

$$J_{CE} = ??$$

Q1

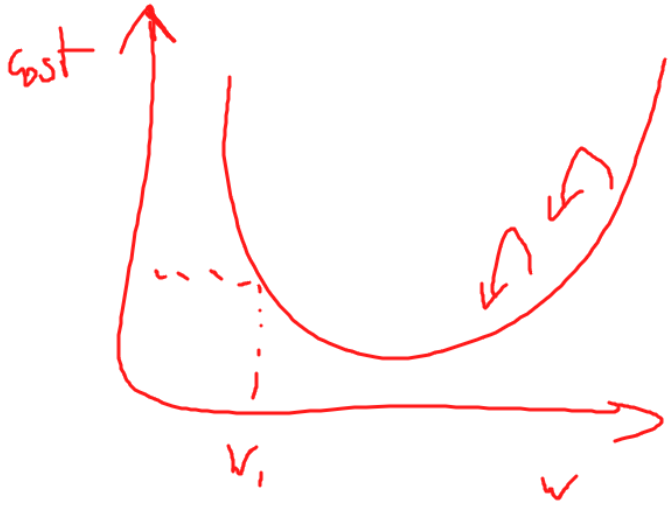
$$\frac{dJ_{CE}}{dw} =$$



* In Linear Regression (LSE or MSE) which is

$$\frac{1}{2}\sum (y - \hat{y})^2$$

$$\hat{y} = wx + b$$





(*) In Logistic Regression we use

Cross Entropy [Binary , Multiclass]

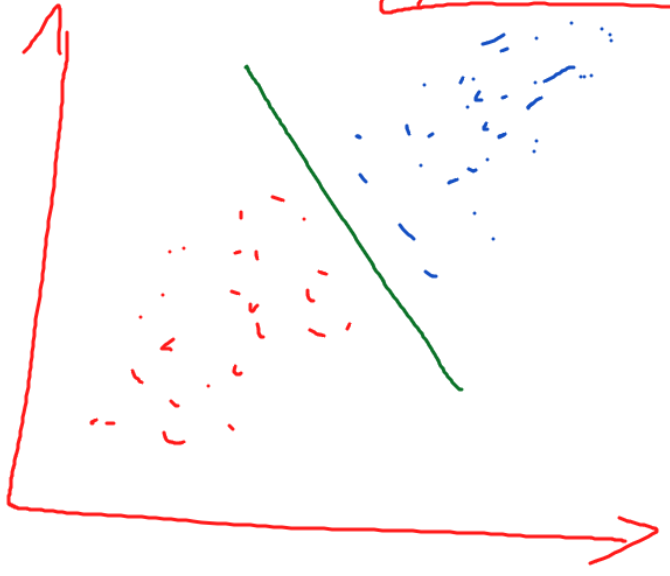
$z = \vec{w}\vec{x} + b$

$\hat{y} = \text{sigmoid}(z)$

Cross entropy







Cross Entropy (binary)

real class

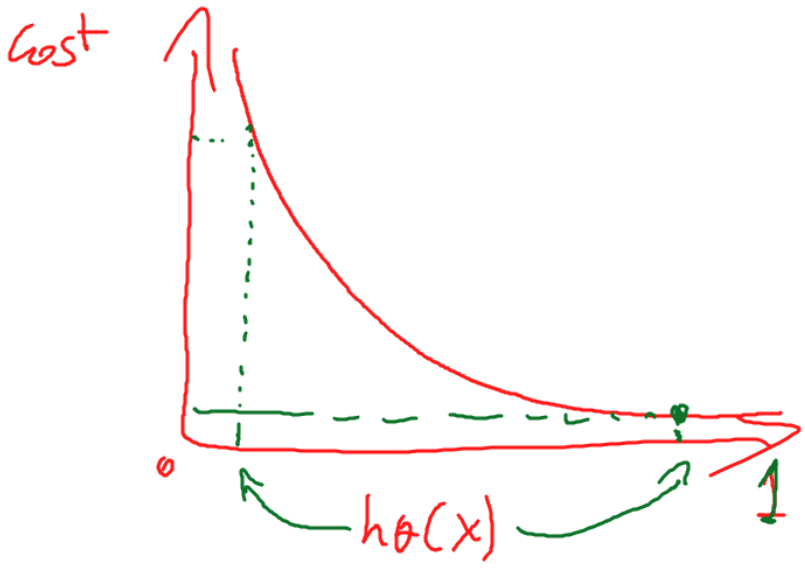
$$J = \begin{cases} -\log\left(g(\vec{w}x+b)\right) \rightarrow \text{if} \quad y=1 \\ -\log\left(1-g(\vec{w}x+b)\right) \rightarrow \text{if} \quad y=0 \end{cases}$$

$$J = y\log\left(g(\vec{w}x+b)\right) + (1-y)\log\left(1-g(\vec{w}x+b)\right)$$

$\theta = (w, b)$

(x, y)

$(1, 1)$ $\quad y = 1$

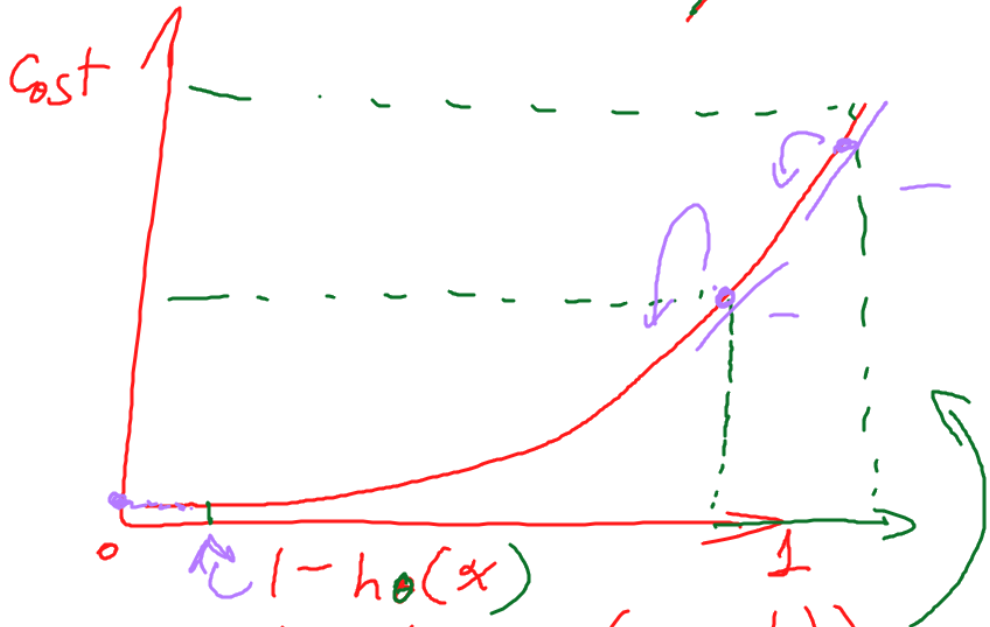


$h_\theta(x) = -\log(\sigma(xw+b))$

gradient descent

(x, y)

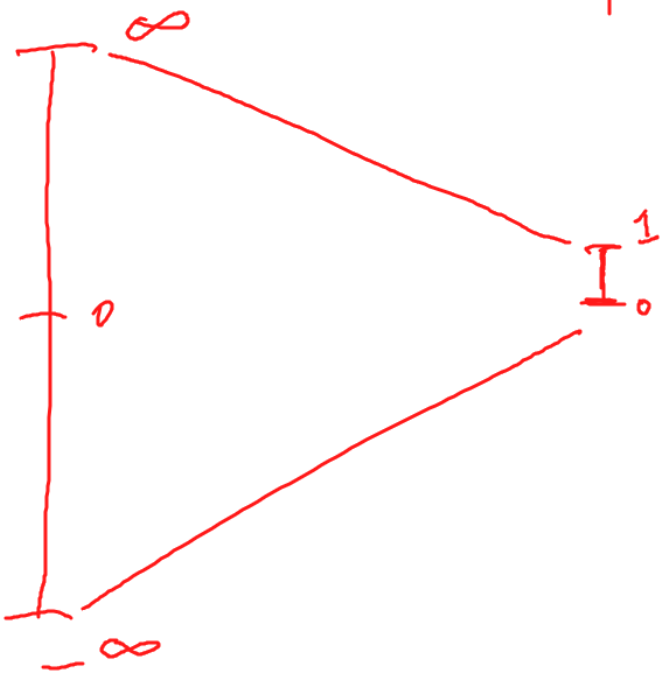
$(0, 0)$ $\quad y = 0$



$1-h_\theta(x) = -\log(1-\sigma(wx+b))$

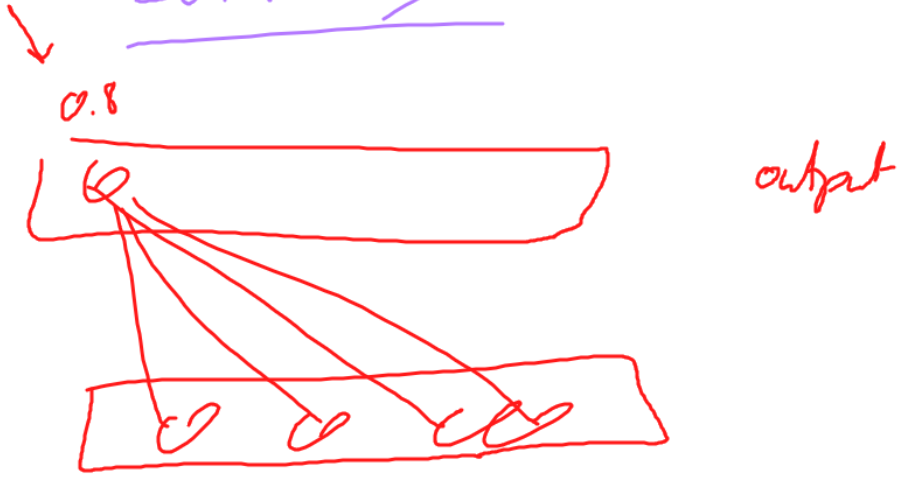
2

$$g(z) = \frac{1}{1 + e^{-z}}$$



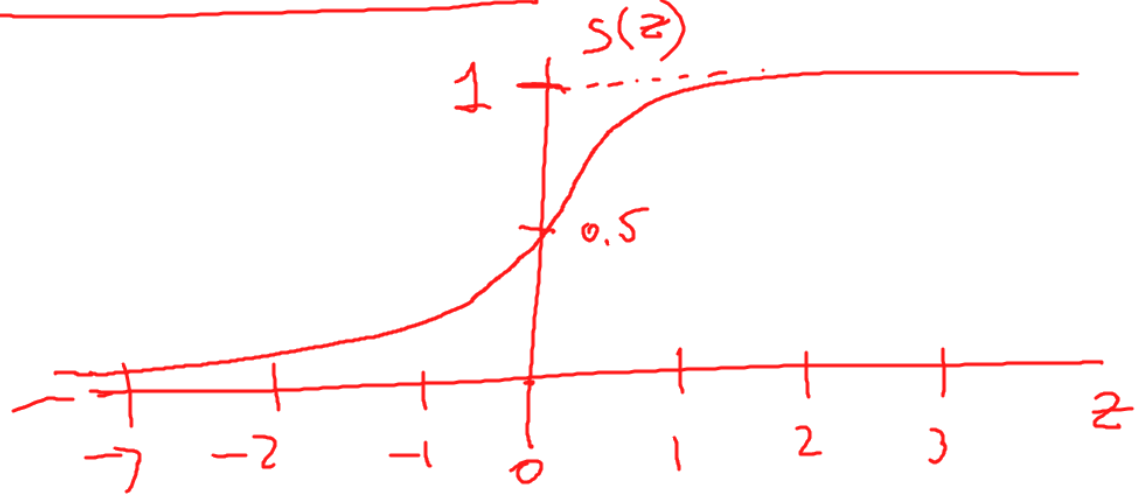
Tensorflow (P.L)

Soft max



Sigmoid:

$$S(z) = \frac{1}{1 + e^{-z}}$$



$$z = wx + b$$

$$\hat{y} = S(z)$$

$$\hat{y} = S(\vec{w}\vec{x} + b)$$

x	ŷ
0	0
0	0
1	1
0	1